



VLSI Lab

CSE 405

Experiment Number: 04

Design and Simulation of an 8-Bit Register Using D Flip-Flops in Using Verilog HDL in Xilinx ISE

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Title of the Experiment

Hierarchical Design and Behavioral Simulation of an 8-Bit Register Utilizing D Flip-Flop Modules in Verilog HDL (Xilinx ISE 14.7 Platform).

Objectives

- To construct an 8-bit synchronous register using a modular, bottom-up design methodology in Verilog HDL.
- To implement a positive-edge-triggered D flip-flop primitive using procedural always blocks and non-blocking assignments.
- To encapsulate the D flip-flop into a 1-bit register wrapper module and instantiate eight parallel units to form an 8-bit register.
- To simulate and verify circuit functionality across diverse 8-bit input data patterns in the ISim behavioral simulation environment.
- To analyze and evaluate synthesized RTL schematics and hierarchical structures generated by Xilinx Synthesis Technology (XST).

Theoretical Background

Sequential Circuits

Sequential logic circuits rely on memory mechanisms where output states are determined by both immediate inputs and historical circuit states. Unlike combinational logic, storage elements such as latches and flip-flops allow sequential systems to maintain state memory over clock cycles.

The positive-edge-triggered D Flip-Flop serves as the standard single-bit storage element in digital system design. It samples the input data (D) exclusively at the rising edge of the clock signal (clk) and holds this value stable at output Q until the subsequent rising edge occurs.

Clock	D	Q (next state)
No rising edge	X	Q (previous / no change)
Rising edge (↑)	0	0
Rising edge (↑)	1	1

By integrating eight 1-bit register blocks driven by a single synchronous clock line, the 8-bit register captures and stores an entire 8-bit word simultaneously on every rising edge. Utilizing non-blocking assignment operators in sequential blocks guarantees concurrent evaluation during hardware modeling, avoiding race conditions.

1-Bit Register

A 1-bit register is constructed by packaging a D flip-flop inside its own module. In its simplest form, as used in this experiment, the register module contains no additional logic beyond the D flip-flop itself; its purpose is to provide a clean, reusable interface that can later be extended with features such as an enable signal or an asynchronous reset without changing the interface seen by higher-level modules.

8-Bit Register

An 8-bit register is formed by placing eight independent 1-bit registers side by side, each storing one bit of an 8-bit data word (D7 D6 D5 D4 D3 D2 D1 D0), and driving all eight with a single, common clock signal. On every rising edge of the clock, all eight bits are captured simultaneously, so the register updates as one atomic 8-bit unit. Registers of this kind are fundamental building blocks in larger digital systems, such as the program counter, instruction register, and accumulator in a simple computer datapath.

Hierarchical (Bottom-Up) Design Methodology

This experiment follows a bottom-up hierarchical design methodology: the D flip-flop is designed first as the leaf-level module, the 1-bit register instantiates the D flip-flop, and the 8-bit register instantiates eight copies of the 1-bit register. This approach improves modularity, makes the design easier to verify and debug one level at a time, and allows the design to be scaled (for example, to a 16-bit or 32-bit register) simply by instantiating more copies of the lower-level module.

Hardware and Software Requirements

- **Development Software:** Xilinx ISE Design Suite 14.7
- **Simulation Environment:** ISim Simulator
- **Synthesis Engine:** Xilinx Synthesis Technology
- **Virtualization Platform:** Oracle VirtualBox
- **Virtual Environment:** Oracle Linux 64-bit (
- **Host Operating System:** Windows 11 / Windows 10
- **Hardware Description Language:** Verilog HDL
- **Version Control & Portfolio Platform:** Git & GitHub Repository

Design Hierarchy and RTL Schematic

The design consists of three levels of modules, summarized below:

Module	Ports	Description
d_flip_flop_v	D, clk (in) Q (out)	Single-bit positive-edge-triggered D flip-flop
register_1bit_v	D, clk (in) Q (out)	Wraps one d_flip_flop_v instance (DFF1)
register_8bit_v	D0–D7, clk (in) Q0–Q7 (out)	Instantiates eight register_1bit_v modules (REG0–REG7) sharing a common clock

At the top level, the RTL schematic generated by XST shows the register_8bit_v block with nine inputs (D0–D7, clk) and eight outputs (Q0–Q7). Expanding the block reveals eight identical register_1bit_v sub-blocks (REG0–REG7) connected in parallel, all sharing the same clk net, and expanding any one of these further reveals the single D flip-flop primitive that XST maps the design onto. This nested view confirms that the synthesis tool interpreted the Verilog hierarchy exactly as intended: one flip-flop per bit, all clocked together.

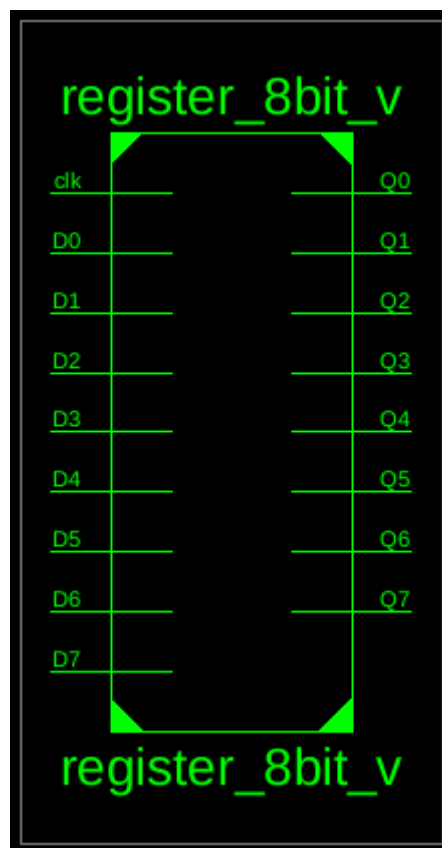


Figure 1: Top-level RTL schematic of the 8-bit register

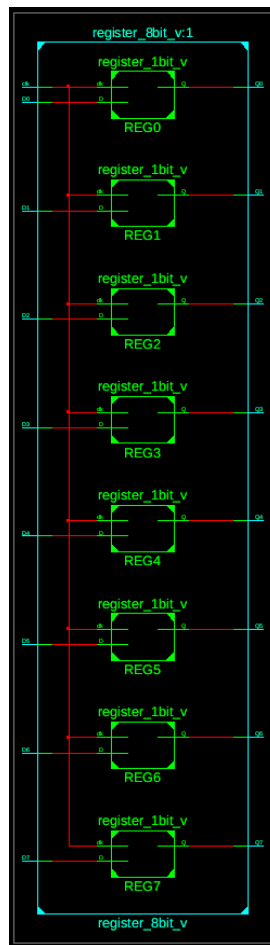


Figure 2: Expanded RTL schematic showing REG0–REG7

Port Define

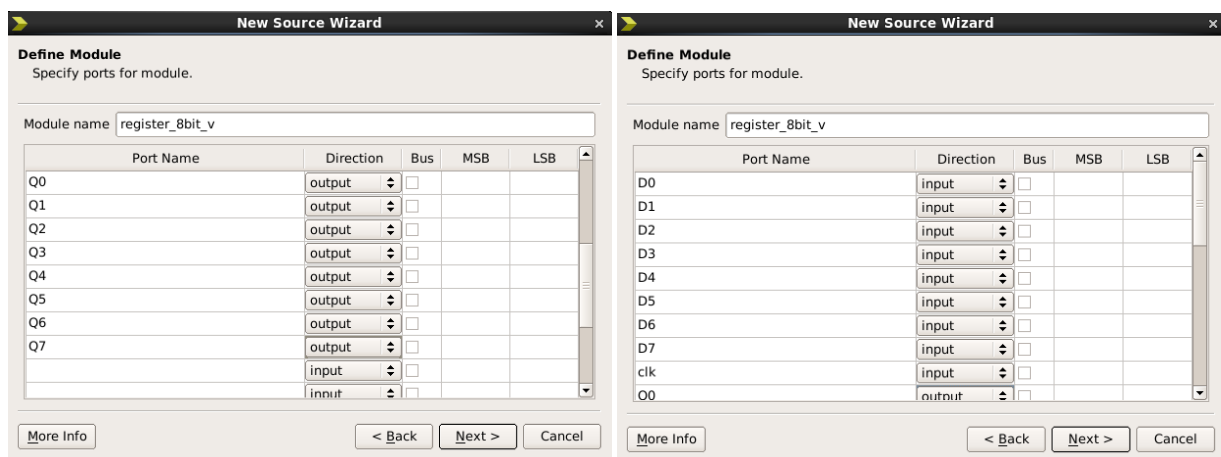


Figure 3: Ports

Simulation and Result

The design was verified using behavioural simulation in the ISim simulator. The testbench (register_8bit_v_tb.v) instantiates register_8bit_v as the unit under test, generates a clock of 10 ns period using an always/forever block (clk toggling every 5 ns), and applies four different 8-

bit data patterns on D0–D7, each held for 10 ns: all zeros (00000000), an alternating pattern (01010101), the complementary alternating pattern (10101010), and all ones (11111111). Because each rising clock edge falls in the middle of a 10 ns data interval, the D inputs are always stable well before they are sampled, so the outputs Q0–Q7 update to match D exactly at each rising edge, with no glitches. Comparing the simulated waveform against the expected behaviour of an edge-triggered register, all four data patterns were captured correctly, confirming that the design is functionally correct.

Time (ns)	D7 D6 D5 D4 D3 D2 D1 D0	Rising Clock Edge	Q7 Q6 Q5 Q4 Q3 Q2 Q1 Q0 (after edge)
0 – 10	0 0 0 0 0 0 0 0	5 ns	0 0 0 0 0 0 0 0
10 – 20	1 0 1 0 1 0 1 0	15 ns	1 0 1 0 1 0 1 0
20 – 30	0 1 0 1 0 1 0 1	25 ns	0 1 0 1 0 1 0 1
30 – 40	1 1 1 1 1 1 1 1	35 ns	1 1 1 1 1 1 1 1

The modular 8-bit register circuit was successfully developed, synthesized, and verified in Xilinx ISE 14.7. The ISim behavioral simulation results confirmed precise edge-aligned state transitions across all tested vectors without output latency or dynamic glitches. RTL synthesis in XST correctly identified and expanded the three-tier hierarchy into eight parallel synchronous D flip-flops driven by a shared clock tree.

The objectives of the experiment were achieved. The D flip-flop was described using an always block sensitive to the positive clock edge, the 1-bit register was built by instantiating the flip-flop, and the 8-bit register was built by instantiating eight 1-bit registers. The testbench verified correct data capture for multiple input patterns, and the RTL schematic was viewed after synthesis.

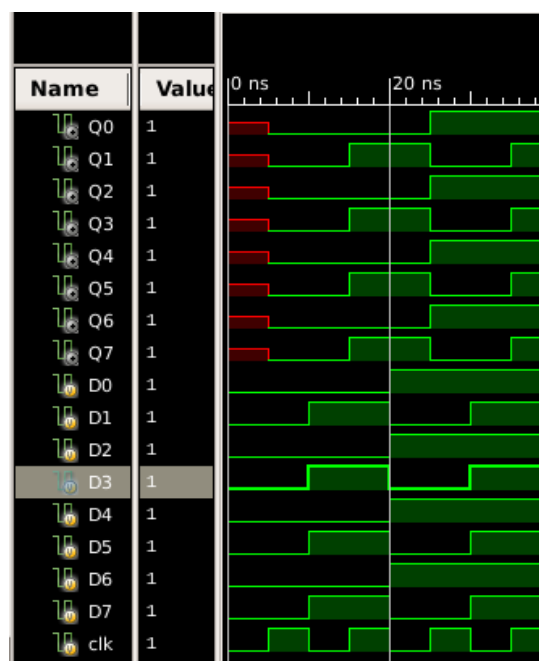


Figure 4: Simulation waveform of the 8-bit register

Verilog Source Code

d_flip_flop_v.v:

```
module d_flip_flop_v(  
    input D,  
    input clk,  
    output reg Q  
);  
    always @(posedge clk) begin  
        Q <= D;  
    end  
endmodule
```

register_1bit_v.v:

```
module register_1bit_v(  
    input D,  
    input clk,  
    output Q  
);  
    d_flip_flop_v DFF1 (  
        .D(D),  
        .clk(clk),  
        .Q(Q)  
    );  
endmodule
```

register_8bit_v.v:

```
module register_8bit_v(  
    input D0, input D1, input D2, input D3,  
    input D4, input D5, input D6, input D7,  
    input clk,  
    output Q0, output Q1, output Q2, output Q3,  
    output Q4, output Q5, output Q6, output Q7  
);  
    register_1bit_v REG0 (.D(D0), .clk(clk), .Q(Q0));  
    register_1bit_v REG1 (.D(D1), .clk(clk), .Q(Q1));  
    register_1bit_v REG2 (.D(D2), .clk(clk), .Q(Q2));  
    register_1bit_v REG3 (.D(D3), .clk(clk), .Q(Q3));  
    register_1bit_v REG4 (.D(D4), .clk(clk), .Q(Q4));  
    register_1bit_v REG5 (.D(D5), .clk(clk), .Q(Q5));  
    register_1bit_v REG6 (.D(D6), .clk(clk), .Q(Q6));  
    register_1bit_v REG7 (.D(D7), .clk(clk), .Q(Q7));  
endmodule
```

register_8bit_v_tb.v:

```
`timescale 1ns / 1ps

module register_8bit_v_tb;
    reg D0, D1, D2, D3, D4, D5, D6, D7;
    reg clk;
    wire Q0, Q1, Q2, Q3, Q4, Q5, Q6, Q7;

    register_8bit_v uut (
        .D0(D0), .D1(D1), .D2(D2), .D3(D3),
        .D4(D4), .D5(D5), .D6(D6), .D7(D7),
        .clk(clk),
        .Q0(Q0), .Q1(Q1), .Q2(Q2), .Q3(Q3),
        .Q4(Q4), .Q5(Q5), .Q6(Q6), .Q7(Q7)
    );

    initial begin
        clk = 0;
        forever #5 clk = ~clk;
    end

    initial begin
        D0=0; D1=0; D2=0; D3=0; D4=0; D5=0; D6=0; D7=0; #10;
        D0=0; D1=1; D2=0; D3=1; D4=0; D5=1; D6=0; D7=1; #10;
        D0=1; D1=0; D2=1; D3=0; D4=1; D5=0; D6=1; D7=0; #10;
        D0=1; D1=1; D2=1; D3=1; D4=1; D5=1; D6=1; D7=1; #10;
        $finish;
    end
end
endmodule
```

GitHub Repository Link

[The completed source code](#)

Conclusion

In this experiment, an 8-bit synchronous register was successfully modeled using a bottom-up hierarchical architecture in Verilog HDL and verified within the Xilinx ISE 14.7 design suite. By instantiating basic D flip-flops within a 1-bit module and arraying eight parallel units under a unified clock tree, modular design scalability was effectively demonstrated. Functional simulation in ISim validated correct data capture on rising clock edges across multiple data vectors. Synthesis via XST generated an accurate RTL structure representing parallel D flip-flop logic primitives. This practical implementation solidifies fundamental concepts of sequential digital systems and HDL hardware modeling.